

Module 10: Fractal Geometry and the Limits of Prediction — Answer Key

Prequiz

- $7 \rightarrow 22 \rightarrow 11 \rightarrow 34 \rightarrow 17 \rightarrow 52 \rightarrow 26 \rightarrow 13 \rightarrow 40 \rightarrow 20 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$.
 There are $\boxed{16}$ steps, and the peak value is $\boxed{52}$.
- At step 3, the number of segments is $4^3 = \boxed{64}$.
 Total length:

$$\left(\frac{4}{3}\right)^3 = \frac{64}{27} \approx \boxed{2.370}$$

 The length factor per step is $\boxed{4/3}$.
- $$\left(\frac{3}{4}\right)^4 = \frac{81}{256} \approx \boxed{0.3164}$$
- $x_1 = 2.8(0.2)(0.8) = \boxed{0.448}$.
 To determine x_2 , substitute $x_1 = 0.448$ into the same recursion with $r = 2.8$.
- Measured lengths are
 $10(50) = 500, \quad 5(130) = 650, \quad 2.5(340) = 850, \quad 1.25(900) = 1125$.
 The measured length increases as the ruler gets shorter because the measurement scale becomes finer and captures more small boundary detail.

Matching quiz

- $6 \rightarrow 3 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$.
 There are $\boxed{8}$ steps, and the peak is $\boxed{16}$.
- Step 4 has $4^4 = \boxed{256}$ segments. Total length is multiplied by

$$\left(\frac{4}{3}\right)^4 = \frac{256}{81} \approx \boxed{3.1605}$$

 from step 0 to step 4.
- $$\left(\frac{3}{4}\right)^3 = \frac{27}{64} = \boxed{0.421875}$$
- $x_1 = 3(0.5)(0.5) = \boxed{0.75}$,
 $x_2 = 3(0.75)(0.25) = \boxed{0.5625}$.
- It demonstrates **scale-dependent measurement**: a finer ruler follows more small-scale boundary detail, so the numerical measurement can change even though the physical coastline has not changed.